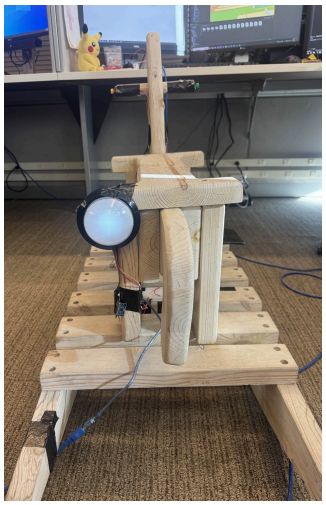
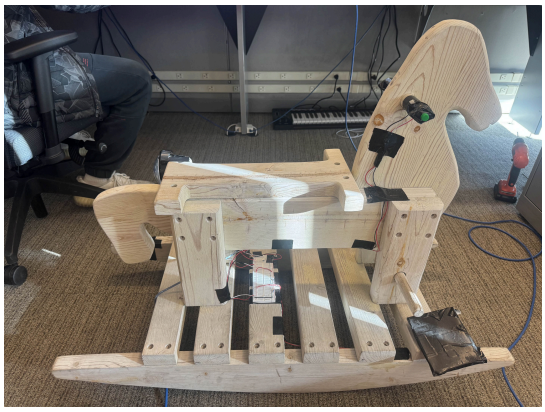
Prototype Dev Diary

This prototype was a great experience for me and something I dedicated a lot of time to because I wanted to continue to the next round. I personally contributed to all the hardware and wiring, from the electronics like programming the Arduino to modifying the physical controller (the rocking horse). The challenge game initially from not understanding why the gyro sensor wasn't outputting data accurately, but after researching and working with other teams, I was able to figure it out. Next working/communicating with the software engineers to integrate the sensor into the game code took some time to figure out, but felt rewarding once it started to work. Through the testing and iteration process, the main issue was the connection of the sensors to the Arduino board. It kept losing connection so planning out how to wire it and place it on the horse was another challenge that I eventually nailed down the night before the presentation. A final challenge that wasn’t as high risk was getting a big arcade button properly wired. There was no clear wiring guide online so I had to experiment with a tone with different connections. When I got the LED working I realized what each contact pin was in charge of. The problem was the sensitivity of the button click made it so the button was reading an input when not pressed. I figured out that through code I could activate the built-in resistor on the button so that it gave the proper resistance to only trigger when the button was clicked. This in turn worked on all buttons, meaning one less wire was needed; however, it was working and I didn’t want to waste time or resources removing it, so I left it as there was no reason to remove it to begin with. Overall I believe with my extra work to ensure a smooth and intuitive experience with gameplay, the end prototype worked well and was fun for those who tested it.





A situation that arose during the final week of the prototype was trying to mount an Arduino board, that we put in a protective case, onto the horse. I told our producer to let me figure it out and that I would have to run to the restroom. They did not hear that and proceeded to drill a screw that got stuck and stripped. The task was now to remove the screw without breaking the casing. It was a high-stress situation due to running low on time, but the Action I took to solve the problem was to not place blame, and instead grab the necessary tools to remove the screw. After 30 minutes we were able to get the screw out and mount the case using command strips. As a result, we had to break the case, but we bought an extra one. I think communication was a lesson learned here, and also not to overstep your role and let the “expert” handle what they were responsible for.

